Linear Algebra

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1 Linear Equations

1.1 Systems of Linear Equations

Linear Function

A function f is **linear** in the variables $x_1, x_2, \ldots, x_n$ if it can be written in the form:

$$f\left(x_1, x_2, \ldots, x_n, \ldots\right) = a_1x_1 + a_2x_2 + \ldots + a_nx_n$$

where $a_1, a_2, \ldots, a_n$ are constants with respect to $x_1, x_2, \ldots, x_n$.

- To put it simply, a function is **linear** in the variables $x_1, x_2, \ldots, x_n$ if it can be written as the sum of constant multiples of $x_1, x_2, \ldots, x_n$. These constant multiples are called the **coefficients**.
- For example, the function $f(x, y) = 2xy^2$ is linear in x but not linear in y.

Linear Equation

A **linear equation** in the variables $x_1, x_2, \ldots, x_n$ can be written in the form:

$$f\left(x_1, x_2, \ldots, x_n, \ldots\right) = b$$

where f is linear in $x_1, x_2, \ldots, x_n$ and b is constant with respect to $x_1, x_2, \ldots, x_n$.

- A **linear equation** in the variables $x_1, x_2, \ldots, x_n$ can be written in the form:

  $$a_1x_1 + a_2x_2 + \ldots + a_nx_n = b$$

  where a_i is the coefficient of the variable x_i for all $1 \leq i \leq n$, and b is a constant.
- For example, the equation $2x + 5y = 6z^2$ is linear in x and y but not linear in z.

System of Linear Equations

A **system of linear equations** (or **linear system**) is a collection of one or more linear equations involving the same set of variables.

A **solution** of a linear system in the variables $x_1, x_2, \ldots, x_n$ is an ordered n-tuple $\left(s_1, s_2, \ldots, s_n\right)$ that, when substituted for $\left(x_1, x_2, \ldots, x_n\right)$ respectively, makes each equation in the system a true statement.

- The **solution set** of a system of equations is the set of all possible solutions for that system.
  - Two systems are **equivalent** if they have the same solution set.
- A linear system can have one solution, no solution, or infinitely many solutions.
  - A linear system is **consistent** if it has at least one solution (i.e., one or infinite)
  - A linear system is **inconsistent** if it has no solutions.

Example: Solving and Classifying a Linear System

Find the solution set to the following system of equations, and classify the system as consistent or inconsistent:

$$\begin{aligned} x_1 - 2x_2 &= -1 \\ -x_1 + 3x_2 &= 3 \end{aligned}$$

Adding the two equations together, we get $x_2 = 2$, which implies $x_1 - 2(2) = -1 \implies x_1 = 3$.

Thus, one solution to the system is $(3, 2)$. Since it is the only solution, the solution set is $\{(3, 2)\}$, and the system is consistent.

Example: Textbook Exercise 1.1.P4

For what values of h and k make the following system consistent?

$$\begin{aligned} 2x_1 - x_2 &= h \\ -6x_1 + 3x_2 &= k \end{aligned}$$

We can multiply the first equation by -3 and add it to the second equation:

$$0x_1 + 0x_2 = k - 3h \implies 0 = k - 3h$$

For the system to be consistent, we need $k - 3h = 0 \implies k = 3h$.

Otherwise, if $k - 3h \neq 0$, the system is inconsistent because there would be no solutions (i.e., a contradiction).

Introduction to Matrices

- A **matrix** is a collection of numbers arranged into a rectangular array of rows and columns.
  - A matrix with m rows and n columns is called an $m \times n$ matrix.
  - If $\mathbf{A}$ is a matrix, then $\mathbf{A}_{ij}$ denotes the entry in the i-th row and j-th column of $\mathbf{A}$.
- To distinguish between scalars, vectors, and matrices, multiple notations are used:
  - Scalars are lowercase and italicized, e.g., a, b, c.
  - Vectors can either be:
    - lowercase and bolded: $\mathbf{v}$, $\mathbf{w}$, or
    - lowercase with an arrow on top: $\vec{v}, \vec{w}$.
    - unit vectors are sometimes denoted with a hat: $\hat{\imath}, \hat{\jmath}, \hat{k}$.
  - Matrices are uppercase and (usually) bolded, e.g., $\mathbf{A}, \mathbf{B}, \mathbf{C}$.
    - sometimes they are simply italicized: A, B, C (as in PDP^{-1})
- A vector $\langle v_1, v_2, v_3, \ldots \rangle$ can be represented as a matrix (or made "compatible" with matrices) by writing it as either a **column matrix** (or **column vector**): $\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \\ \vdots \end{pmatrix}$

  or a **row matrix** (or **row vector**): $\mathbf{v} = \begin{pmatrix} v_1 & v_2 & v_3 & \cdots \end{pmatrix}$
- The **coefficient matrix** of a linear system is the matrix consisting of the coefficients of the variables in the system. For example, the system:

$$\begin{aligned} x_1 - 2x_2 + x_3 &= 0 \\ 2x_2 - 8x_3 &= 8 \\ 5x_1 - 5x_3 &= 10 \end{aligned}$$

can be represented by the coefficient matrix $\begin{pmatrix} 1 & -2 & 1 \\ 0 & 2 & -8 \\ 5 & 0 & -5 \end{pmatrix}$.

- The **augmented matrix** of a linear system is the matrix consisting of the coefficient matrix with an additional column that contains the constants from the right-hand side of each equation.

  For example, the augmented matrix for the above system is $\begin{pmatrix} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ 5 & 0 & -5 & 10 \end{pmatrix}$.

Solving Linear Systems

One strategy to solve a linear system is to perform **elementary row operations** on the associated augmented matrix, eliminating variables one at a time until we arrive at a trivial form.

An example of a trivial form is the **triangular form**, where all entries below the main diagonal are zero. Below is a template for a 3×4 triangular matrix:

$$\begin{pmatrix} * & * & * & * \\ 0 & * & * & * \\ 0 & 0 & * & * \end{pmatrix}$$

The main diagonal is highlighted in blue.

Elementary Row Operations

There are three **elementary row operations** which operate on individual rows of a matrix (which can correspond to equations in a linear system):

1. *Swap* two rows $R_i \leftrightarrow R_j$
2. *Multiply* by a nonzero scalar: $R_i \to kR_i$ where $k \neq 0$
3. *Add* a multiple of one row to another: $R_i \to R_i + kR_j$

Note: Operation 3 is sometimes $R_i \to aR_i + bR_j$

- Row operations can be applied to any matrix.
- Two matrices are **row equivalent** if there exists a sequence of elementary row operations that transforms one matrix into the other.
  - If the augmented matrices of two linear systems are row equivalent, then the two systems are equivalent (i.e., have the same solution set).
- Row operations are *reversible*, and the following lists inverse operations:
  - $R_i \leftrightarrow R_j$ is its own inverse.
  - $R_i \to kR_i$ is inverted by $R_i \to \frac{1}{k}R_i$.
  - $R_i \to R_i + kR_j$ is inverted by $R_i \to R_i - kR_j$.

Example: Solving a Linear System

Solve the following system of equations:

$$\begin{aligned} x_1 - 2x_2 + x_3 &= 0 \\ 2x_2 - 8x_3 &= 8 \\ 5x_1 - 5x_3 &= 10 \end{aligned}$$

We can use the augmented matrix to represent the system, and label the rows:

$$\begin{array}{cc} \begin{pmatrix} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ 5 & 0 & -5 & 10 \end{pmatrix} & \begin{matrix} R_1 \\ R_2 \\ R_3 \end{matrix} \end{array}$$

Start with the equation with the most leading zeros, which is R_2. We can use this to eliminate x_2 from R_1 by adding R_2 to R_1 ($R_1 \to R_1 + R_2$):

$$\begin{array}{cc} \begin{pmatrix} 1 & 0 & -7 & 8 \\ 0 & 2 & -8 & 8 \\ 5 & 0 & -5 & 10 \end{pmatrix} & \begin{matrix} R_1 \\ R_2 \\ R_3 \end{matrix} \end{array}$$

Notice that only R_1 changed, and that we targeted R_1 because it had the most nonzero entries.

Next, we can use R_1 to eliminate x_1 from R_3 by performing the operation $R_3 \to R_3 - 5R_1$:

$$\begin{array}{cc} \begin{pmatrix} 1 & 0 & -7 & 8 \\ 0 & 2 & -8 & 8 \\ 0 & 0 & 30 & -30 \end{pmatrix} & \begin{matrix} R_1 \\ R_2 \\ R_3 \end{matrix} \end{array}$$

Finally, we can use R_3 to eliminate x_3 from R_2 by performing the operation $R_2 \to R_2 + \frac{4}{15}R_3$:

$$\begin{array}{cc} \begin{pmatrix} 1 & 0 & -7 & 8 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 30 & -30 \end{pmatrix} & \begin{matrix} R_1 \\ R_2 \\ R_3 \end{matrix} \end{array}$$

We have achieved triangular form, so we can start back-substitution. Here, we see the final matrix corresponds to the system of equations:

$$\begin{aligned} x_1 - 7x_3 &= 8 \\ 2x_2 &= 0 & &\implies x_2 = 0 \\ 30x_3 &= -30 & &\implies x_3 = -1 \end{aligned}$$

Finally, $x_1 - 7(-1) = 8 \implies x_1 + 7 = 8 \implies x_1 = 1$. Thus, the solution is $(1, 0, -1)$.

1.2 Row Reduction & Echelon Forms

Echelon Form

A rectangular matrix is in **echelon form** or **row echelon form** (REF) if it is:
1. *Separated*: nonzero rows are above any rows of all zeros.
2. *Ordered*: the leading entry (or **pivot**) of each nonzero row is to the right of the leading entry of the previous row.
3. *Normalized Below*: all entries below a leading entry are zeros.

The leading entry (or **pivot**) of a nonzero row is the first nonzero entry from the left in that row.

A matrix is in **reduced row echelon form** (or **row reduced echelon form**, RREF) if it is in echelon form and also meets the following additional properties:
1. *Leading 1s*: the leading entry in each nonzero row is 1.
2. *Normalized Everywhere*: all entries above and below a leading 1 are zeros.

- A matrix in echelon form must be in triangular form, but the converse is not necessarily true.
- A matrix in echelon form is called an **echelon matrix**, and a matrix in RREF is called a **reduced echelon matrix**.

Existence and Uniqueness of Reduced Row Echelon Form

Every matrix is row equivalent to one and only one reduced echelon matrix.
In other words, each matrix has ***exactly one*** *reduced row echelon form.*

Pivots and Free Variables

- The **pivot** positions of a matrix are the locations of the leading entries in its nonzero rows.
  - The columns that contain pivot positions are called **pivot columns**.
  - The number of pivot positions is called the **rank** of the matrix.
- The columns that do not contain pivot positions are called **free columns**.
  - The variables corresponding to free columns are called **free variables**.
    As their name suggests, these variables can take on any value in the solution set.
  - The other variables (variables in pivot columns) are called **basic variables**.
- Basic variables may be expressed in terms of the free variables in a linear system. This way of expressing the solution set is called the **parametric description** (or form) of the solution set.
  - The following are two equivalent parametric descriptions:

$$\left\{ \left(x_1, x_1 + 2\right) \,\middle|\, x_1 \in \mathbb{R} \right\} \equiv \begin{cases} x_1 \text{ is free} \\ x_2 = x_1 + 2 \end{cases}$$

Existence Theorem

Let A be the augmented matrix of a linear system. If the rightmost column of A is <u>not</u> a pivot column, then the linear system must be consistent (and there must exist at least one solution).

- The Existence Theorem is intuitive to understand. The only way to have the rightmost column be a pivot is to have a row in the form $(0, 0, \dots, b)$ with $b \neq 0$. This corresponds to the equation $0 = b$, which is a contradiction and thus makes the system inconsistent.

> **Uniqueness Theorem**
> Let A be the augmented matrix of a *consistent* linear system. If every variable in the system is a basic variable (i.e., there are no free variables), then the linear system has a unique solution.

The Row Reduction Algorithm

Let's consider the following matrix:

$$\begin{pmatrix} 1 & 2 & -1 & -4 \\ 2 & 3 & 3 & 3 \\ 4 & 5 & 1 & 6 \end{pmatrix}$$

To reduce this matrix into **echelon form** (ref), we can follow these steps (the **forward phase**):

1. Choose the leftmost column as the **pivot column**. Our goal with a pivot column is to turn it into a column with a pivot position in the first row. That is, we want to have a leading entry in the first row and zeros below it.

$$\begin{pmatrix} \color{red}{1} & 2 & -1 & -4 \\ \color{blue}{2} & 3 & 3 & 3 \\ \color{blue}{4} & 5 & 1 & 6 \end{pmatrix}$$

The pivot column is highlighted in blue, and the pivot position is highlighted in red.
Only focusing on the blue column, we want to use row operations to make everything below the red entry equal to zero.

2. If necessary, swap rows to move a nonzero entry into the pivot position (first row of the pivot column). In this case, the first entry is already nonzero, so we can skip this step.

3. Use row operations to create zeros below the pivot position. We can eliminate the entries below the pivot by performing the following row operations:

$$R_2 \to R_2 - 2R_1 \qquad \begin{pmatrix} \color{red}{1} & 2 & -1 & -4 \\ \color{blue}{0} & -1 & 5 & 11 \\ \color{blue}{4} & 5 & 1 & 6 \end{pmatrix}$$

$$R_3 \to R_3 - 4R_1 \qquad \begin{pmatrix} \color{red}{1} & 2 & -1 & -4 \\ \color{blue}{0} & -1 & 5 & 11 \\ \color{blue}{0} & -3 & 5 & 22 \end{pmatrix}$$

4. Now, consider the pivot column as the next column to the right (that contains a nonzero entry). The row of our pivot position will also move down one row.

   In this case, we can use the second column as the next pivot column:

$$\begin{pmatrix} 1 & \color{blue}{2} & -1 & -4 \\ 0 & \color{red}{-1} & 5 & 11 \\ 0 & \color{blue}{-3} & 5 & 22 \end{pmatrix}$$

   Performing row operations to create the zero below the pivot position:

$$R_3 \to R_3 - 3R_2 \qquad \begin{pmatrix} 1 & 2 & -1 & -4 \\ 0 & -1 & 5 & 11 \\ 0 & 0 & -10 & -11 \end{pmatrix}$$

5. Finally, we can use the third column as the next pivot column:

$$\begin{pmatrix} 1 & 2 & -1 & -4 \\ 0 & -1 & 5 & 11 \\ 0 & 0 & -10 & -11 \end{pmatrix}$$

   Since there are no entries below the pivot position, we have achieved echelon form (REF).

To convert the matrix into **reduced row echelon form** (RREF), we can follow these steps (the **backward phase**):

1. Starting from the *last* pivot column, use row operations to create zeros *above* the pivot position. Starting with the third column:

$$R_2 \to R_2 + \frac{1}{2}R_3 \qquad \begin{pmatrix} 1 & 2 & -1 & -4 \\ 0 & -1 & 0 & 5.5 \\ 0 & 0 & -10 & -11 \end{pmatrix}$$

$$R_1 \to R_1 - \frac{1}{10}R_3 \qquad \begin{pmatrix} 1 & 2 & 0 & -2.9 \\ 0 & -1 & 0 & 5.5 \\ 0 & 0 & -10 & -11 \end{pmatrix}$$

2. If the pivot is not 1, scale accordingly:

$$R_3 \to -\frac{1}{10}R_3 \qquad \begin{pmatrix} 1 & 2 & 0 & -2.9 \\ 0 & -1 & 0 & 5.5 \\ 0 & 0 & 1 & 1.1 \end{pmatrix}$$

3. Repeat steps 1 and 2 for the remaining pivot columns, moving leftward:

$$R_1 \to R_1 + 2R_2 \qquad \begin{pmatrix} 1 & 0 & 0 & 8.1 \\ 0 & -1 & 0 & 5.5 \\ 0 & 0 & 1 & 1.1 \end{pmatrix}$$

$$R_2 \to -1R_2 \qquad \begin{pmatrix} 1 & 0 & 0 & 8.1 \\ 0 & 1 & 0 & -5.5 \\ 0 & 0 & 1 & 1.1 \end{pmatrix}$$

$$\text{left column is good} \qquad \begin{pmatrix} 1 & 0 & 0 & 8.1 \\ 0 & 1 & 0 & -5.5 \\ 0 & 0 & 1 & 1.1 \end{pmatrix}$$

This is now in reduced row echelon form (RREF), and we can read off the solution directly:

$$(x_1, x_2, x_3) = (8.1, -5.5, 1.1).$$

1.3 Vector Equations

Span of a Set of Vectors

Let $V = \{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\} \subseteq \mathbb{R}^n$. The **span** of V, denoted $\operatorname{span}(V)$ or $\operatorname{span}\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}$, is the set of all linear combinations of the vectors in V:

$$\operatorname{span}(V) = \{\vec{v} \in \mathbb{R}^n \mid \vec{v} = c_1\vec{v}_1 + c_2\vec{v}_2 + \dots + c_n\vec{v}_n \text{ for } c_1, c_2, \dots, c_n \in \mathbb{R}\}.$$

It holds that $\operatorname{span}(V)$ is a subspace of $\mathbb{R}^n$.

1.4 The Matrix Equation $A\mathbf{x} = \mathbf{b}$

1.5 Solution Sets of Linear Systems

- A system of linear equations is **homogeneous** if it can be written in the form $A\mathbf{x} = \mathbf{0}$, where A is the coefficient matrix and $\mathbf{0}$ is the zero vector.
  - A homogeneous system is *always* consistent because the **trivial solution** $\mathbf{x} = \mathbf{0}$ always satisfies the equation.
- For homogeneous systems, we are often interested in finding the **nontrivial solutions**.

Existence of a Nontrivial Solution

Let A be the coefficient matrix of a homogeneous linear system. The system has a nontrivial solution if and only if the equation $A\mathbf{x} = \mathbf{0}$ has at least one free variable.

1. Let $\mathbf{x} = \mathbf{x}_h$ solve $A\mathbf{x} = \mathbf{0}$. Then, for any constant $c \in \mathbb{R}$, $\mathbf{x} = c\mathbf{x}_h$ is also a solution to $A\mathbf{x} = \mathbf{0}$.
2. If $\mathbf{x} = \mathbf{p}$ solves $A\mathbf{x} = \mathbf{b}$, then the solution set of $A\mathbf{x} = \mathbf{b}$ is given by:

$$\{\mathbf{x} = \mathbf{p} + \mathbf{x}_h \mid A\mathbf{x}_h = \mathbf{0}\}$$

1.6 Applications of Linear Systems

1.7 Linear Independence

1.8 Linear Transformations

1.9 The Matrix of a Linear Transformation

2 Matrix Algebra

2.1 Matrix Operations

Matrix Multiplication

Let $\mathbf{A}$ be an $m \times p$ matrix and $\mathbf{B}$ be an $p \times n$ matrix. The **product AB** is the $m \times n$ matrix where the entry in the i^{th} row and j^{th} column is given by:

$$(\mathbf{AB})_{ij} = \sum_{k=1}^{p} A_{ik} B_{kj}.$$

AB represents the transformation that results from first applying the transformation represented by **B**, followed by the transformation represented by **A**.

- In other words, to compute the entry in the i^{th} row and j^{th} column of the product matrix **AB**, we take the dot product of the i^{th} row of **A** with the j^{th} column of **B**:

$$(\mathbf{AB})_{ij} = \vec{A}_{i*} \cdot \vec{B}_{*j}.$$

Matrix Transpose

The **transpose** of an $m \times n$ matrix $\mathbf{A}$, denoted $\mathbf{A}^\mathsf{T}$, is the $n \times m$ matrix obtained by interchanging the rows and columns of $\mathbf{A}$. In other words, the entry in the i^{th} row and j^{th} column of $\mathbf{A}^\mathsf{T}$ is equal to the entry in the j^{th} row and i^{th} column of $\mathbf{A}$:

$$\left(\mathbf{A}^\mathsf{T}\right)_{ij} = \mathbf{A}_{ji}.$$

2.2 The Inverse of a Matrix

2.3 Invertible Matrices

Invertible Matrix Theorem

Let A be an $n \times n$ matrix. The following statements are equivalent (i.e. all true or all false):

1. A and/or $A^\top$ is invertible.
2. The set of rows in A is equal to the set of rows in I_n.
3. A has n pivot positions.
4. The equation $A\vec{x} = \vec{0}$ only has the solution $\vec{x} = \vec{0}$.
5. The columns of A are linearly independent.
6. The linear transformation $\vec{x} \mapsto A\vec{x}$ is one-to-one and/or onto (in/sur/bijective).
7. The equation $A\vec{x} = \vec{b}$ has at least one solution for each $\vec{b}$ in $\mathbb{R}^n$.
8. The columns of A span $\mathbb{R}^n$.
9. There exists an $n \times n$ matrix C such that $CA = I_n$.
10. There exists an $n \times n$ matrix D such that $AD = I_n$.

2.4 Partitioned Matrices

2.5 Matrix Factorizations

LU Factorization

2.6 The Leontief Input-Output Model

2.7 Subspaces of $\mathbb{R}^n$

2.8 Dimension and Rank

Dimension of a Subspace

The **dimension** of a subspace $\mathbf{W}$ of $\mathbb{R}^n$ is the number of vectors in a basis for $\mathbf{W}$. We denote the dimension of $\mathbf{W}$ as $\dim(\mathbf{W})$.

Rank of a Matrix

The **rank** of a matrix A, denoted $\operatorname{rank}(A)$, is the dimension of the column space of A (or equivalently, the row space of A).

3 Determinants

3.1 Introduction to Determinants

- The **determinant** of a square matrix is a scalar value which represents the factor by which the area/volume of a region is scaled when the linear transformation associated with the matrix is applied to that region.
- If the determinant of a matrix is zero, the linear transformation compresses the area/volume to zero and we cannot reverse the transformation (i.e., the matrix is not invertible).

Determinant of a Matrix

The **determinant** ($\det A$ or $|A|$) of an $n \times n$ matrix A ($n \geq 2$) is:

$$\det A = \sum_{j=1}^{n} (-1)^{1+j} A_{1j} \det\left(M_{1j}\right)$$

where M_{1j} is the minor of A at row 1, column j, which is the $(n-1) \times (n-1)$ matrix that results from deleting row 1 and column j from A

- The determinant of a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is $ad - bc$. (We can write $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$.)

Cofactor Expansion

We can actually expand along any row or column, not just the first row:

$$\det A = \sum_{j=1}^{n} (-1)^{i+j} A_{ij} \det\left(M_{ij}\right) \qquad \text{expansion along } i^{\text{th}} \text{ row}$$

$$\det A = \sum_{i=1}^{n} (-1)^{i+j} A_{ij} \det\left(M_{ij}\right) \qquad \text{expansion along } j^{\text{th}} \text{ column}$$

All expansions will yield the same determinant. Notice the sign $(-1)^{i+j}$ follows a checkerboard pattern, allowing us to easily determine whether to add or subtract each term:

$$\begin{pmatrix} + & - & + & - \\ - & + & - & + \\ + & - & + & - \\ - & + & - & + \end{pmatrix}$$

The **cofactor** of an entry A_{ij} is defined as $C_{ij} = (-1)^{i+j} \det\left(M_{ij}\right)$, capturing what we multiply by A_{ij} in the determinant expansion. The **cofactor matrix** C of an $n \times n$ matrix A is the matrix where each entry C_{ij} is the cofactor of A_{ij}.

Computing determinants using cofactors is known as the method of **cofactor expansion**.

Example: Using the Cofactor Expansion

Show that the cofactor expansion along the first row of a 3×3 matrix results in the same determinant as the expansion along the second column.

Let $A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$. The cofactor expansion along the first row is:

$$\begin{aligned} \det A &= a\begin{vmatrix} e & f \\ h & i \end{vmatrix} - b\begin{vmatrix} d & f \\ g & i \end{vmatrix} + c\begin{vmatrix} d & e \\ g & h \end{vmatrix} \\ &= a(ei - fh) - b(di - fg) + c(dh - eg) \\ &= aei - afh - bdi + bfg + cdh - ceg. \end{aligned}$$

The cofactor expansion along the second column is:

$$\begin{aligned} \det A &= -b\begin{vmatrix} d & f \\ g & i \end{vmatrix} + e\begin{vmatrix} a & c \\ g & i \end{vmatrix} - h\begin{vmatrix} a & c \\ d & f \end{vmatrix} \\ &= -b(di - fg) + e(ai - cg) - h(af - cd) \\ &= -bdi + bfg + eai - ecg - haf + hcd \\ &= aei - afh - bdi + bfg + cdh - ceg. \end{aligned}$$

Thus, both expansions yield the same determinant.

3.2 Properties of Determinants

- The three elementary row operations affect the determinant as follows:
  1. Swapping two rows multiplies the determinant by -1.
  2. Multiplying a row by a scalar k multiplies the determinant by k.
  3. Adding a multiple of one row to another row does not change the determinant.
- $\det A = \det A^{\mathsf{T}}$

3.3 Cramer's Rule

Cramer's Rule

Let A be an invertible $n \times n$ matrix, and $\mathbf{b} \in \mathbb{R}^n$. Then the unique solution $\mathbf{x}$ to the equation $A\mathbf{x} = \mathbf{b}$ has entries given by:

$$x_i = \frac{\det A_i}{\det A} \text{ for } i = 1, 2, \dots, n$$

where A_i is the matrix that results from replacing the i^{th} column of A with the column vector $\mathbf{b}$, or in other words:

$$(A_i)_{jk} = \begin{cases} b_j & \text{if } k = i \\ A_{jk} & \text{otherwise} \end{cases}$$

- To put it simply, Cramer's Rule allows us to solve for a particular variable in a system of linear equations.
  1. Take the determinant of the coefficient matrix A, this is the denominator ($\det A$).
  2. Replace the i^{th} column of A with the constants from the right-hand side of the equations to form A_i, and take its determinant, this is the numerator ($\det A_i$).
  3. The solution for the variable whose coefficients were in the i^{th} column (x_i) is $x_i = \frac{\det A_i}{\det A}$.
- Realize that the coefficient matrix must be square, i.e. we must have the same number of equations as unknowns.
  - Additionally, the coefficient matrix must be invertible (i.e., $\det A \neq 0$) for Cramer's Rule to be applicable. Otherwise, the system may have no solutions or infinitely many solutions.
- Cramer's Rule is most efficient for solving one variable in large systems, but may not be the best method for solving for all variables (since it involves calculating $n + 1$ determinants).

4 Vector Subspaces

5 Eigeneverything

5.1 Eigenvectors and Eigenvalues

Eigenvector and Eigenvalue

Let $\mathbf{A}$ be an $n \times n$ matrix. A nonzero vector $\mathbf{v} \in \mathbb{R}^n$ is an **eigenvector** of $\mathbf{A}$ if there exists a scalar λ such that:

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}.$$

The scalar λ is called the **eigenvalue** corresponding to the eigenvector $\mathbf{v}$. $\mathbf{A}$ may have multiple pairs of eigenvectors and eigenvalues.

5.2 The Characteristic Equation

Derivation: Characteristic Equation

Let $\mathbf{A}$ be an $n \times n$ matrix, and let $\mathbf{v}$ be an eigenvector of $\mathbf{A}$ with corresponding eigenvalue λ. Then:

1. By the definitions of eigenvector and eigenvalue, $\mathbf{Av} = \lambda\mathbf{v}$.
2. Rearranging gives $\mathbf{Av} - \lambda\mathbf{v} = \mathbf{0}$.
3. It can be shown that for a column vector $\mathbf{v}$ and scalar k, $k\mathbf{v} = k\mathbf{Iv}$, where $\mathbf{I} = \mathbf{I}_n$ is the $n \times n$ identity matrix.
4. Thus, $\mathbf{Av} - \lambda\mathbf{Iv} = \mathbf{0}$, and by the distributive property, we have $(\mathbf{A} - \lambda\mathbf{I})\mathbf{v} = \mathbf{0}$.
5. Since $\mathbf{v}$ is nonzero, by the Invertible Matrix Theorem, $\mathbf{A} - \lambda\mathbf{I}$ must be invertible and $\det(\mathbf{A} - \lambda\mathbf{I}) = 0$.

5.3 Diagonalization

$$\mathbf{A} = \mathbf{PDP}^{-1}$$

Diagonalization of a Matrix

An $n \times n$ matrix $\mathbf{A}$ is **diagonalizable** if there exists an invertible matrix $\mathbf{P}$ and a diagonal matrix $\mathbf{D}$ such that:

$$\mathbf{A} = \mathbf{PDP}^{-1}$$

where, given λ_i is the eigenvalue for the i^{th} eigenvector $\mathbf{v}_i$ of $\mathbf{A}$:

$$\mathbf{P} = \begin{pmatrix} \vdots & \vdots & \cdots & \vdots \\ \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_n \\ \vdots & \vdots & \cdots & \vdots \end{pmatrix} \qquad \mathbf{D} = \begin{pmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{pmatrix}.$$

The columns of $\mathbf{P}$ are the eigenvectors of $\mathbf{A}$, and the entries on the main diagonal of $\mathbf{D}$ are the corresponding eigenvalues.

For example, if $\mathbf{A}$ has eigenvalues $\lambda_1 = 3, \lambda_2 = 2, \lambda_3 = 5$ with corresponding eigenvectors $\vec{v}_1 = \langle 1, 0, 0 \rangle, \vec{v}_2 = \langle 0, 1, 0 \rangle, \vec{v}_3 = \langle 0, 0, 1 \rangle$ respectively, then:

$$\mathbf{P} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \qquad \mathbf{D} = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{pmatrix}$$

5.4 Eigenvectors in Linear Transformations

5.5 Complex Eigenvalues

5.6 Discrete Dynamical Systems